

# DATA PREPROCESSING REPORT

**Project :** Multi-Class Skin Disease Classification and Diagnosis Platform

**Dataset :** HAM10000 Skin Lesion Dataset

**Total Images :** 10,015

## 1. Dataset Overview

The HAM10000 dataset is a publicly available dermoscopic image dataset containing 10,015 skin lesion images collected from different patients. The dataset consists of seven different skin disease categories including both benign and malignant lesions. The images are captured using dermatoscopy, providing high-quality visual information suitable for deep learning-based medical image classification. The dataset contains images with varying lesion sizes, colors, shapes, and textures. Before model training, preprocessing is essential to standardize the images, improve quality, and prepare them for deep learning architectures.

### Dataset Summary

| Property          | Value     |
|-------------------|-----------|
| Dataset           | HAM10000  |
| Total Images      | 10015     |
| Disease Classes   | 7         |
| Image Format      | JPG       |
| Image Size        | 128 × 128 |
| Color Mode        | RGB       |
| Training Images   | 7009      |
| Validation Images | 1503      |
| Testing Images    | 1503      |

## 2. Data Preprocessing Steps

The preprocessing pipeline ensures that every image has a consistent format before entering the neural network. The following preprocessing operations were performed:

- Loaded metadata from HAM10000\_metadata.csv.
- Mapped image paths from both image folders.
- Converted disease labels into numerical labels.
- Loaded RGB images using TensorFlow.

- Resized every image to  $128 \times 128$  pixels.
- Converted images into floating-point tensors.
- Normalized pixel values by dividing each pixel by 255.
- Created batches using ImageDataGenerator.
- Generated train, validation and testing datasets.
- Verified dataset integrity before training.

### 3. Data Augmentation

To improve model generalization and reduce overfitting, multiple image augmentation techniques were applied during training. These augmentations generate different variations of existing images while preserving disease characteristics. The augmentation techniques include horizontal flipping, vertical flipping, random rotation, zooming, brightness adjustment, and random cropping. These operations enable the model to become more robust to variations in illumination, orientation, and lesion positioning.

| Technique       | Purpose                    |
|-----------------|----------------------------|
| Horizontal Flip | Improve Generalization     |
| Vertical Flip   | Increase Robustness        |
| Rotation        | Handle Orientation Changes |
| Zoom            | Scale Variations           |
| Brightness      | Lighting Robustness        |

### 4. Handling Class Imbalance

The HAM10000 dataset contains an imbalanced distribution of disease classes. Certain diseases such as Melanocytic Nevus contain thousands of images, whereas Dermatofibroma and Vascular Lesion contain relatively few samples. To overcome this issue, class weights were computed using the `compute_class_weight()` function from Scikit-learn. During model training, larger weights were assigned to minority classes so that the neural network paid more attention to underrepresented diseases. This helps improve fairness and overall classification performance.

### 5. Complete Preprocessing Pipeline

HAM10000 Dataset ↓ Metadata Loading ↓ Image Path Mapping ↓ Label Encoding ↓ Resize Images ( $128 \times 128$ ) ↓ Pixel Normalization ↓ Data Augmentation ↓ Train / Validation / Test Split ↓ Class Weight Calculation ↓ Ready for Deep Learning Models

### 6. Conclusion

The preprocessing stage plays an essential role in achieving reliable skin disease classification performance. Image resizing, normalization, augmentation, and class balancing improve the quality of the training data and enable the deep learning models to learn more discriminative features. The final processed dataset provides a standardized and robust input for CNN and transfer learning models, contributing to improved classification accuracy and better generalization on unseen skin lesion images.